

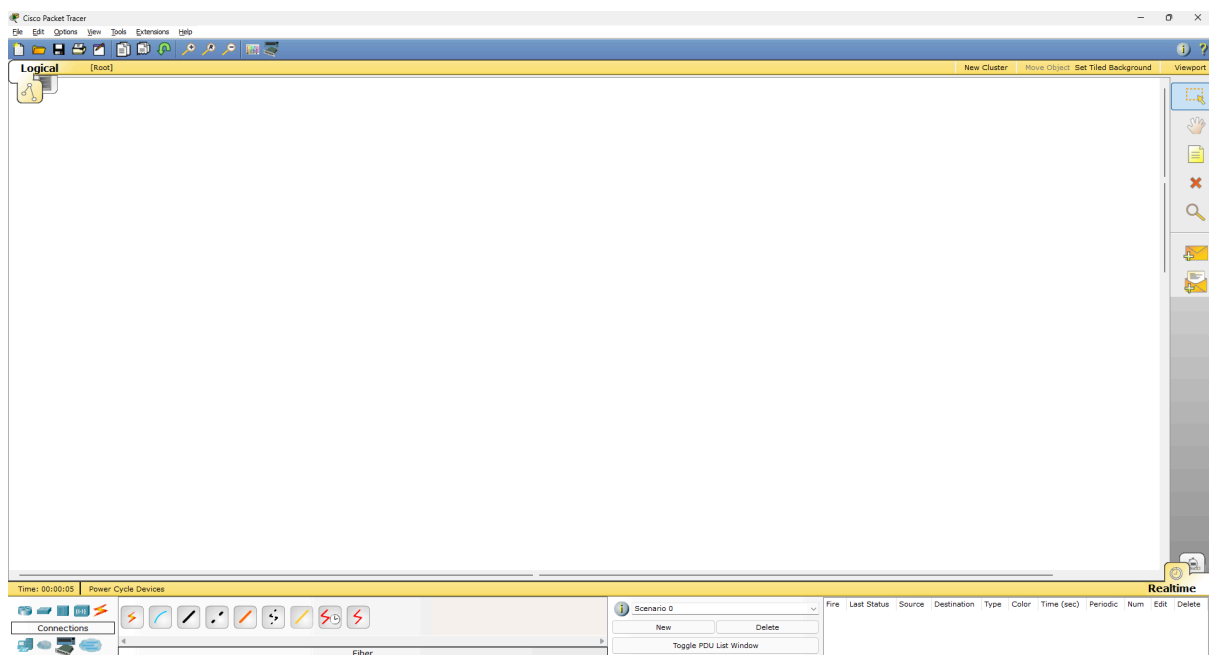
PRACTICALS

Computer Networks Laboratory

Practical 5.

Aim: To implement the RIP routing protocol in the given network having atleast 4 routers.

Step 1: Open Cisco packet tracer 5.1 as shown in FIG 1

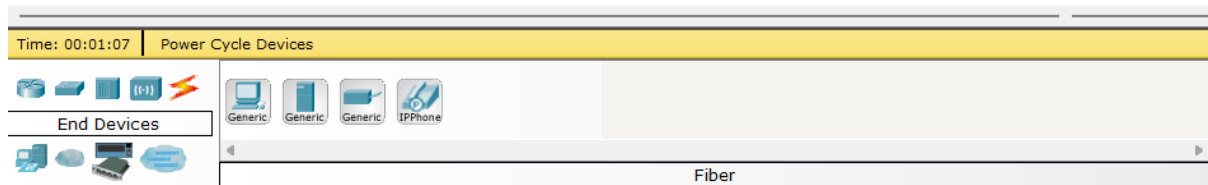


[FIG 1]

Step 2: Select device to make LAN (Local Area Network) Network form the bottom Tool bar as shown in **FIG 2**

There are multiple devices used in this process are follows as:

1. Network Switch
2. Router
3. Hub
4. Wireless device
5. Cables/wires (for connection)
6. End devices (PC, Laptop, Server etc)
7. WAN Emulation
8. Custom made device
9. Multi User connection



[FIG 2]

Step 3: Select PCs, Network Switch, & connecting wires/Cables Shown in **FIG 3**

1. 3 - PCs
2. 1 - Network Switch
3. 3 - Connecting wires/ cables

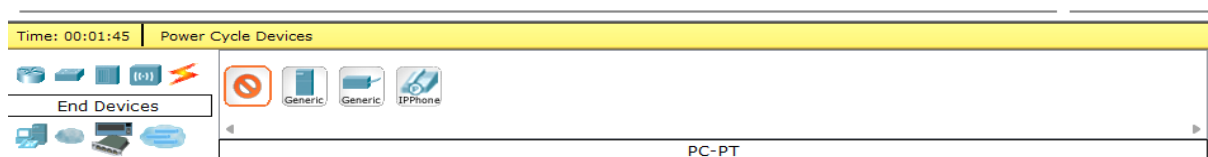


FIG 3

Step 4: Creating Network (Topology) Using selected devices as shown in FIG 4

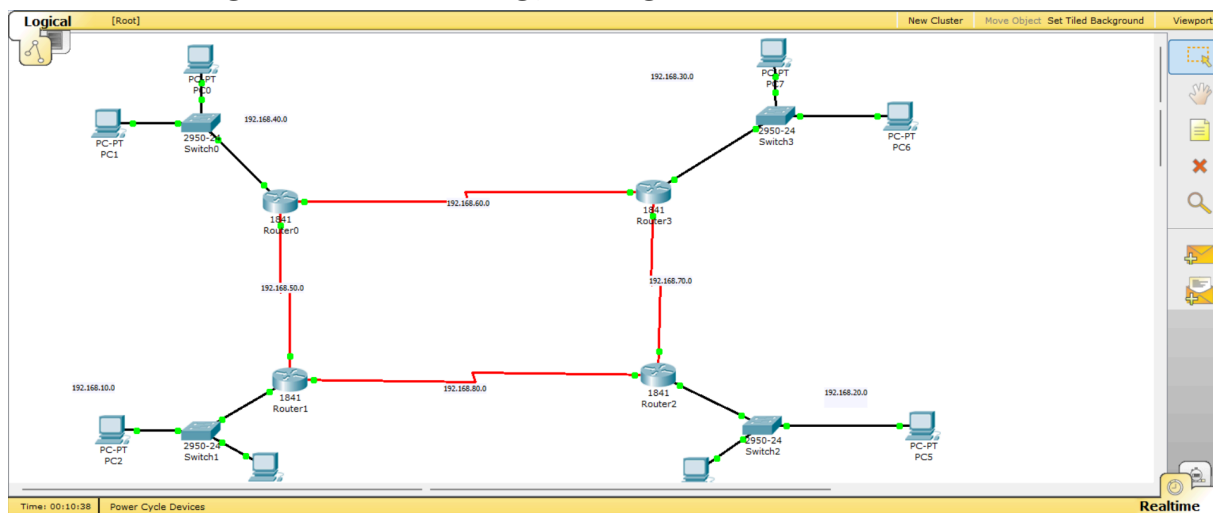


FIG 4

Step 5: Configuring The Setup with Network (static method) Using Default Gateway, Subnet Mask, Ip Address as shown in FIG 5 and FIG 6

1. Click on PC
2. Go to Desktop Tab
3. Click on Ip Configuration

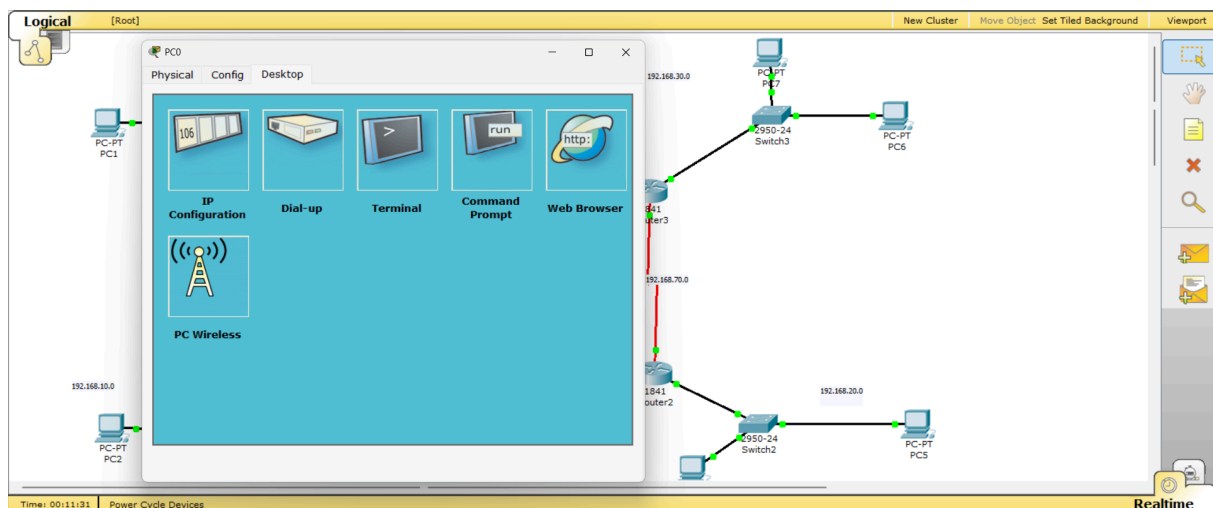


FIG 5

1. Select Or click on Static
2. Enter values as follows
3. Close the dialog box
4. Do the same process for all of the PCs

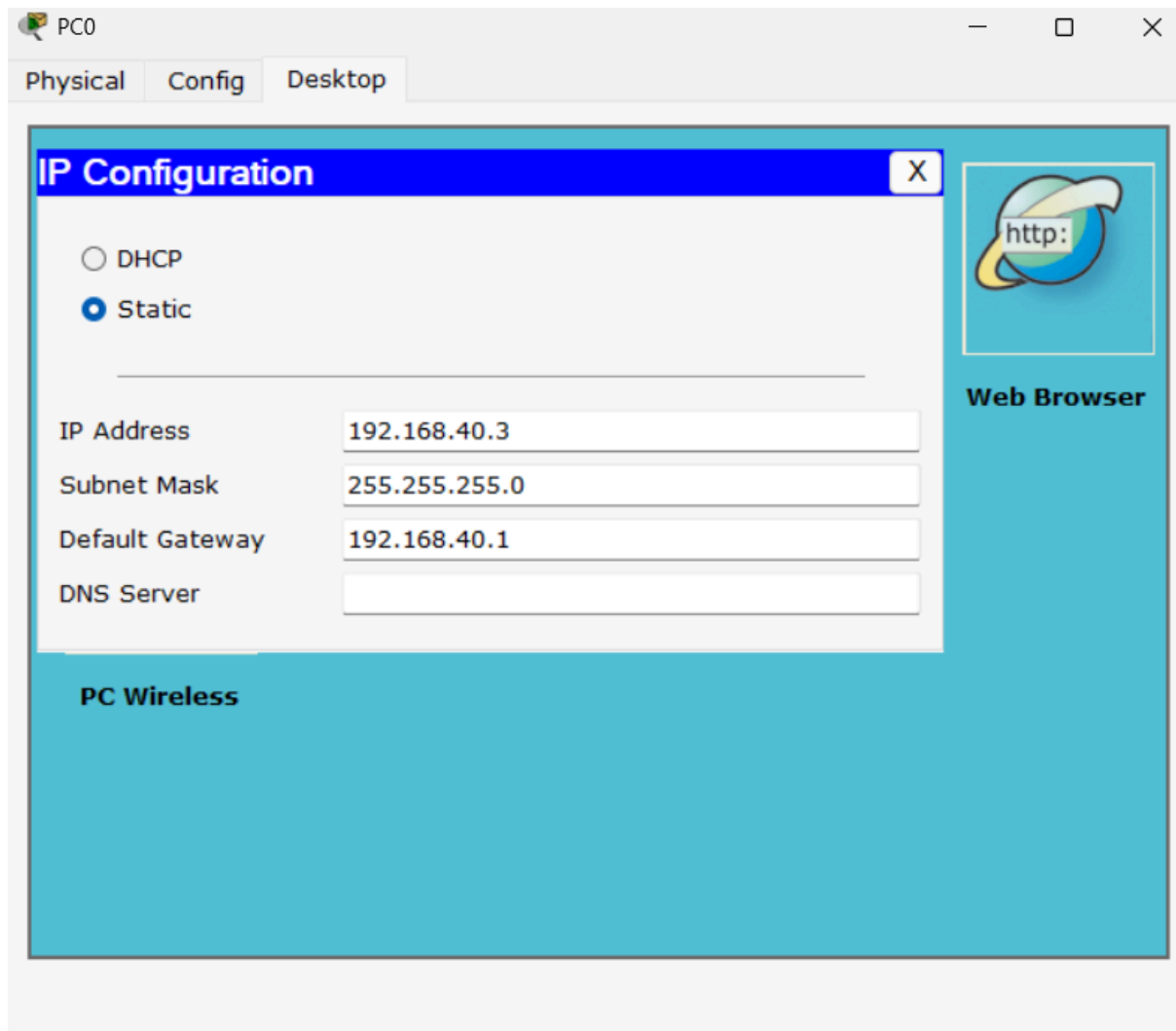


FIG 6

Note:- IP for all pcs are as follows (with subnet mask)

Common Subnet mask for all Pcs:- 255.255.255.0

PC 0,1:- 192.168.40.2, 192.168.40.3

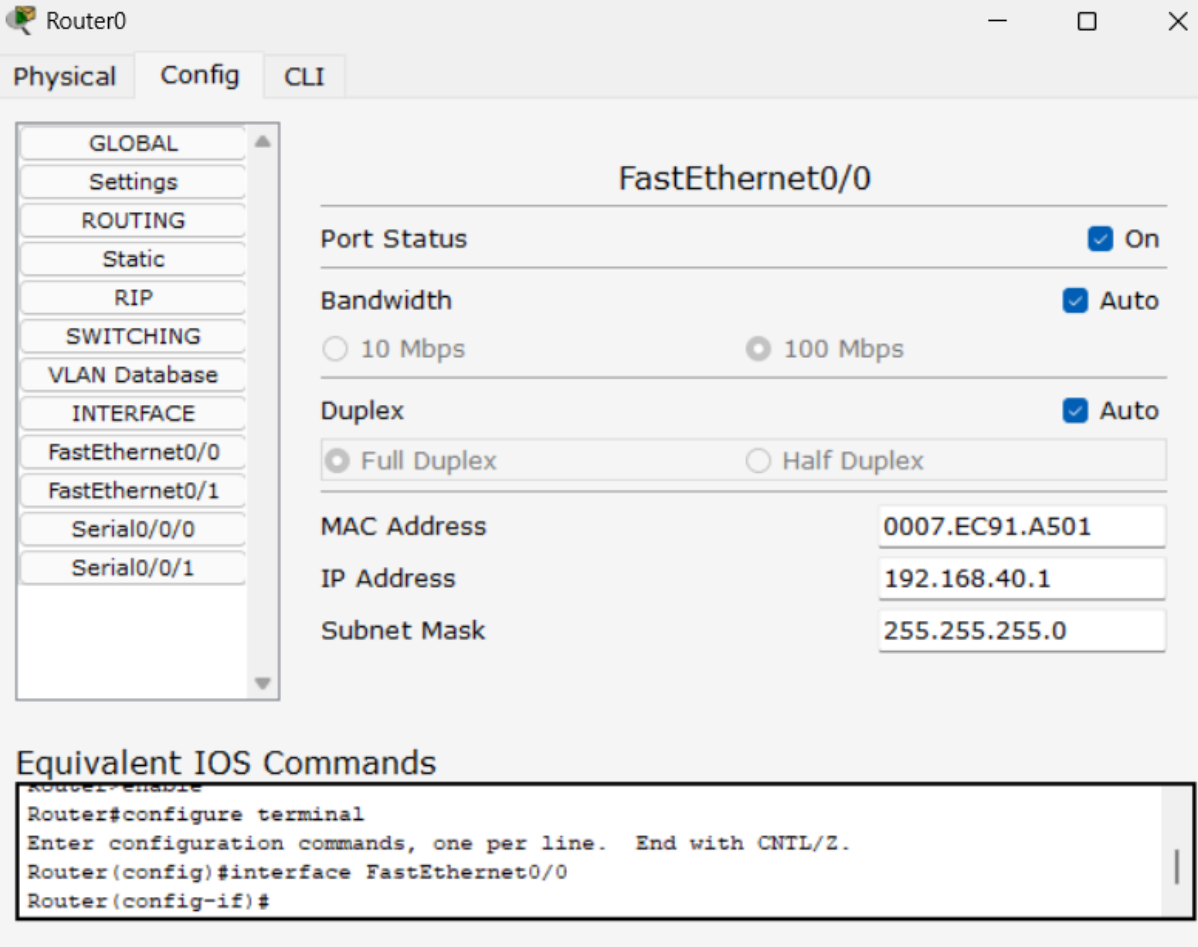
PC 2,3:- 192.168.10.2, 192.168.10.3,

PC 4,5:- 192.168.20.2, 192.168.20.3

PC 6,7:- 192.168.30.2, 192.168.30.3

Step 6: Now Configuring Routers with following steps:

1. Click on Router
2. Click on Config tab
3. Click on FastEthernet0/0 for setup router and network switch
4. Click on serial0/0/0 & serial0/0/0 for setup routers

Step 7: setup the values given in below FIG 7 for FastEthernet 0/0

The screenshot shows the configuration window for Router0 in Cisco Packet Tracer. The 'Config' tab is selected, and the 'FastEthernet0/0' interface is chosen from the left-hand menu. The configuration settings for this interface are as follows:

FastEthernet0/0	
Port Status	☒ On
Bandwidth	☒ Auto
☐ 10 Mbps ☒ 100 Mbps	
Duplex	☒ Auto
☒ Full Duplex ☐ Half Duplex	
MAC Address	0007.EC91.A501
IP Address	192.168.40.1
Subnet Mask	255.255.255.0

Below the configuration settings, the 'Equivalent IOS Commands' are listed in a terminal window:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
```

FIG 7

Step 8: setup the values given in below FIG 8 for serial0/0/0 & FIG 9 for serial0/0/1

The screenshot shows the configuration window for Router0 in Cisco Packet Tracer. The 'Config' tab is selected, and the 'Serial0/0/0' interface is chosen from the left-hand menu. The configuration details for Serial0/0/0 are as follows:

- Port Status:** ☒ On
- Clock Rate:** 64000
- Duplex:** ☐ Full Duplex
- IP Address:** 192.168.50.3
- Subnet Mask:** 255.255.255.0

Below the configuration fields, the 'Equivalent IOS Commands' section displays the following commands:

```
Router(config)#interface Serial0/0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/0
Router(config-if)#
```

FIG 8

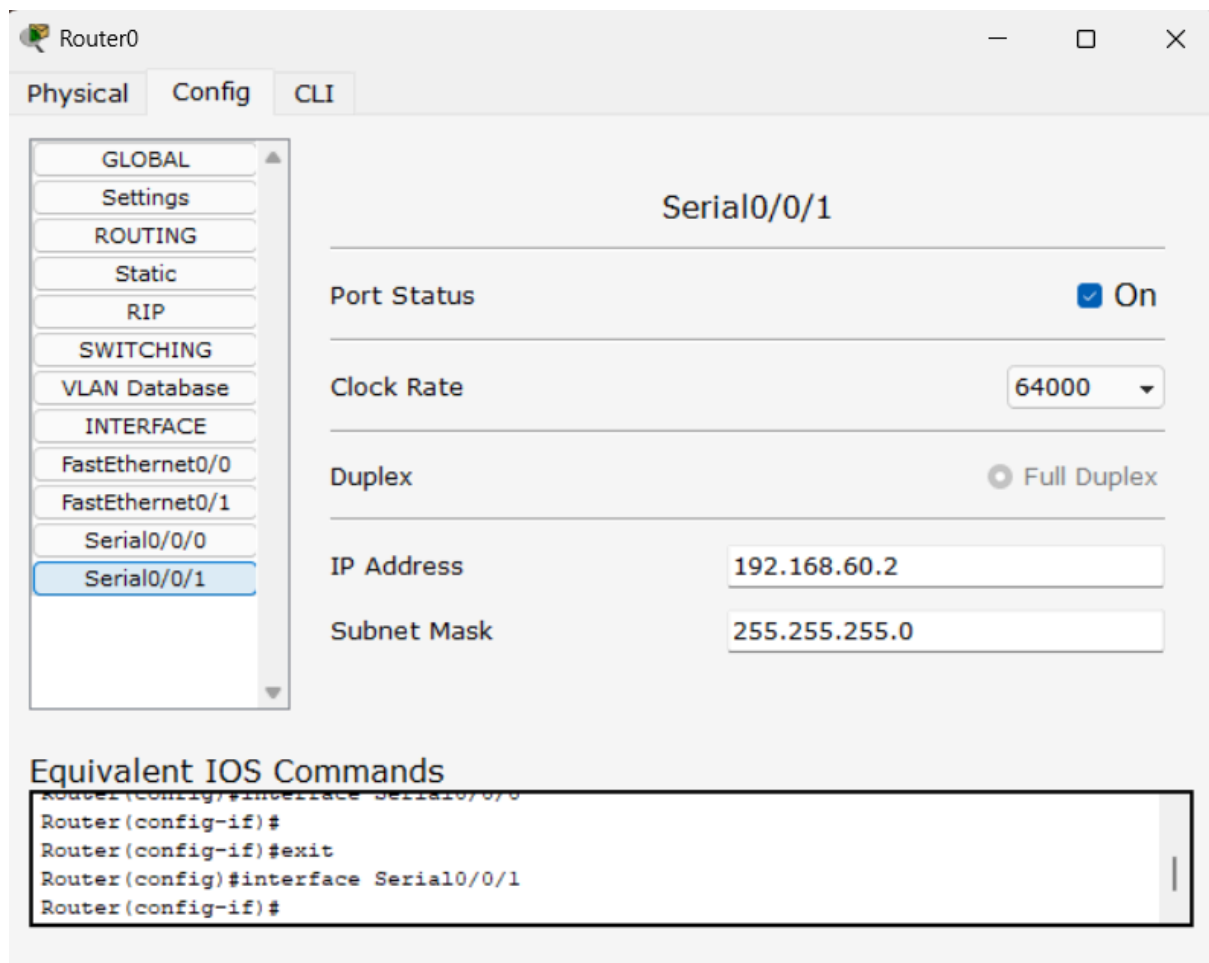


FIG 9

Note:- For other router refer the following table

Router 0

	FastEathernet0/0	Serial0/0/0	Serial0/0/1
Port Status	ON	ON	ON
Clock Rate	–	64000	64000
IP ADDRESS	192.168.40.1	192.168.50.3	192.168.60.2
SUBNET MASK	255.255.255.0	255.255.255.0	255.255.255.0

Router 1

	FastEathernet0/0	Serial0/0/0	Serial0/0/1
Port Status	ON	ON	ON
Clock Rate	–	64000	64000
IP ADDRESS	192.168.10.1	192.168.50.2	192.168.80.2
SUBNET MASK	255.255.255.0	255.255.255.0	255.255.255.0

Router 2

	FastEathernet0/0	Serial0/0/0	Serial0/0/1
Port Status	ON	ON	ON
Clock Rate	–	64000	64000
IP ADDRESS	192.168.20.1	192.168.70.3	192.168.80.3
SUBNET MASK	255.255.255.0	255.255.255.0	255.255.255.0

Router 3

	FastEthernet0/0	Serial0/0/0	Serial0/0/1
Port Status	ON	ON	ON
Clock Rate	—	64000	64000
IP ADDRESS	192.168.30.1	192.168.70.2	192.168.60.3
SUBNET MASK	255.255.255.0	255.255.255.0	255.255.255.0

Step 9: Go to Router and click on Config and goto RIP and add network as shown in FIG 10 for router 0

Router0

Physical Config CLI

GLOBAL

Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/0

FastEthernet0/1

Serial0/0/0

Serial0/0/1

RIP Routing

Network 192.168.40.0

Add

Network Address

192.168.40.0

192.168.50.0

192.168.60.0

Remove

Equivalent IOS Commands

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#
```

FIG 10

Note: Follow the following table for all routers

Router 0
192.168.40.0 192.168.50.0 192.168.60.0

Router 1
192.168.10.0 192.168.50.0 192.168.80.0

Router 2
192.168.20.0 192.168.70.0 192.168.80.0

Router 3
192.168.30.0 192.168.60.0 192.168.70.0

Step 10: Try to send Packets Through-out the network

At first attempt it will failed (to share outside the network) but in second attempt of sending packets it will success as shown in **FIG 11**

Realtime											
 Scenario 0 New Delete Toggle PDU List Window	Fire	Last Status	Source	Destination	Type	Color	Time (sec)	Periodic	Num	Edit	Delete
		Failed	PC0	PC3	ICMP		0.000	N	0	(edit)	(delete)
		Successful	PC0	PC3	ICMP		0.000	N	1	(edit)	(delete)

FIG 11